Using mefpp:
the WFI preprocessing engine

(fselman@eso.org)

eclipse: 5.0.0
qfits: 5.4.0
mefpp: 1.5.0

Introduction

The mefpp pre-processing engine is written in C and based on the Eclipse and qfits software package. It modifies extensively the Eclipse wfi package. It is still at a very early stage of development and should be used carefully. In this document I give an example reduction session so that a user can get started quickly.

The main characteristic of this package is that it maintains the integrity of the Multi Extension FITS files without splitting them into their 8 extensions. Given the complexity of data management when “reducing” astronomical data it was felt that a factor of 8 in the number of files is too high a price to pay.

The program can be used to create calibration files and then reduce the data in one go, or to first produce the calibration data, and only then reduce the science data with the calibrations previously determined. We will follow the last approach in this tutorial.

The current version is based in eclipse-5.0.0, qfits-5.4.0, amd mefpp-1.5.0. Everythin is distributed together as a tar file.

Installation instructions

The program has to be compiled in three sequential stages. Although the main directory created after un-tarring the file is called mefpp-1.5, it really is the top directory of eclipse-5.0.0. Inside this directory there two subdirectories of interest: qfits, and ins/mef.

- Compile qfits:
  - cd into `${MEFPP}`/qfits and run `./configure –prefix=${MEFPP}`
  - then type: `make`;
  - when the compilation ends, type: `make install`
- Compile eclipse:
  - cd `${MEFPP}`
  - run `./configure –prefix=${MEFPP}`
  - run `make`;
  - run `make install`
  - `run: e_setup`
- Compile mefpp:
  - `cd ${MEFPP}/ins/mef`
  - run `make`

At runtime one could get an error about maximum number of pointer allocated. If that happens I suggest to edit the file xmemory.c en qfits/src and change the line:

```
#define XMEMORY_MAXPTRS    200000
```

to a larger value (multiple of 2).

Quick summary for the impatient

Step -1: Copy wfi.paf into working directory as mefpp.paf. Source the file .eclipse-rc which should have been created when mefpp was installed. Make sure that verbose is larger than 0.

Step 0: Explore data with dfits and fitsort

Step 1: Partition data by creating ascii lists using dfits and fitsort

Step 2: Produce calibrations:

```
$mefpp --bias bias.list --biassave --dome domeff_Rc.list --domesave \
--skyff skyff_RC.list --skysave
```

which outputs into working dir:

WFI_MB_1x1.fits
WFI_DOMEFF_BB_Rc_162_ESO844_1x1.fits
WFI_SKYFF_BB_Rc_162_ESO844_1x1.fits
WFI_MFF_BB_Rc_162_ESO844_1x1.fits

Step 3: reduce science frames listed in science.list

```
$mefpp --bias WFI_MB_1x1.fits --mff WFI_SKYFF_BB_Rc_162_ESO844_1x1.fits \
--science science.list
```

Manual Page

mefpp: This program accepts the following switches

--bias <bias.list> or bias.fits

where <bias.list> is a list of Dark frames to be used to calculate a bias frame. You can also give the name of a previously determined bias fits frame, bias.fits. If <bias.list> is given the calculated frame will not be saved unless –biassave is used.

--biassave

if this switch is present it indicates to mefpp to store the bias frame i(f bias.list is given)

--bpm <bpm.list>

where <bpm.list is list of FITS files to be used to calculate a bad pixel mask. These are usually dome flats with different intensities that will be used in a regression analysis pixel by pixel. If omitted the mefpp pre-processing engine requires that you have the BPM files in the working directory: WFI_BPM_01.fits, ..., WFI_BPM_08.fits.

--dome <dome.list> or dome.fits

<dome.list> or dome flat field fits frame to be used in the pre-processing. If <dome.list> is given the fits files contained in the list will be used to calculate a dome flat field frame.

--domesave

include this switch to save the calculated dome flat field, otherwise it will only be used during the mefpp run and will be lost at the end of the processing

--sky <sky.list> or sky.fits

<sky.list> of sky flat field frame to be used in the pre-processing. If <sky.list> is given the FITS files contained in the list will be used to calculate the sky flat field frame.

--skysave

include this switch to save the sky file (important for WFI frames)

--mff mff.fits

master flat field frame name to store the resulting frame of combining sky and dome flats, or if a fits file with this name already exist in disk use it for the pre-processing. This last case is what we use at WFI were we specify the previously saved sky flat field as our reduction master flat field.

--science <science.list>

<science.list> is a file containing the science FITS files that will be pre-processed.

Step-by-step guide

We will assume that you have several nights of data in disk some directory that we will call **rawdir**. Let us assume that the data are store at /data/raw: Our **rawdir** can be any of

/data/raw/2005-07-03 /data/raw/2005-07-04 /data/raw/2005-07-05

We will also assume that you have a directory where you will carry out the reductions, the **workdir**. In what follow it will assumed that you will be issuing commands from a terminal from **workdir.**

Step -1: Copy the mefpp.paf file into your workdir. You can copy either wfi.paf or omegacam.paf. Note that if this file is not present, the program will crash with some criptic message suh as "Segmentation Fault" or "Bus error." It should be called mefpp.paf. It contains all the parameters for the appropriate instrument:

$ cp wfi.paf <workdir>/mefpp.paf

Set the eclipse configuration file .eclipse-rc, which should have ben writen into the home directory when the software was installed, to output processing messages to the console. The file .eclipse-rc in my system reads:

$cat ~/.eclipse-rc

```
#
# eclipse configuration file
#


# Add eclipse to path
E_PATH='/Users/fselman/compu/eso/mefpp' ; export E_PATH
PATH="$PATH:$E_PATH/bin"

# Add verbose/debug
E_VERBOSE=2 ; export E_VERBOSE
E_DEBUG=2 ; export E_DEBUG
# Temporary swap directory
E_TMPDIR=. ; export E_TMPDIR

# end of file
```

Note that E_VERBOSE is set to a value different than 0. This file should be sourced from the same console where you will issue the commands below.

Step 0: Explore the data with **dfits** and **fitsort**. To explore the data it is usefule to be aware that the image files from WFI are stored in Multi-Extension FITS format files, henceforth MEF files. Each of these files consists in a primary header without any image data associated to it, and a sequence of 8 extensions with their own headers and data. Each FITS header consist of a set of 80-character lines with the following format:

KEYWORD = VALUE / Comments

The following is an excerpt of a primary header of a WFI MEF file:

```
SIMPLE  =                    T / Standard FITS format (NOST-100.0)
BITPIX  =                  -32 / Bits per pixel
NAXIS   =                    0 / # of axes in frame
EXTEND  =                    T / Extension may be present.
ORIGIN  = 'ESO     '           / European Southern Observatory
DATE    = '2005-07-05T01:45:28.053'  / UT date when this file was written
MJD-OBS =      53556.06976567 / MJD start (2005-07-05T01:40:27.754)
DATE-OBS= '2005-07-05T01:40:27.754'  / Date of observation
EXPTIME =           299.9178 / Total integration time
OBJECT  = '9PTempel1'        / Target description
INSTRUME= 'WFI     '         / Instrument used
OBSERVER= 'UNKNOWN '         / Name of observer
PI-COI  = 'UNKNOWN '         / Name(s) of proposer(s)
TELESCOP= 'MPI-2.2 '         / ESO Telescope Name
RA      =         204.913936 / 13:39:39.3 RA (J2000) at start in (deg
DEC     =           -9.99834 / -09:59:54.0 DEC (J2000) at start in (d
EQUINOX =               2000. / Standard FK5 (years)
RADECSYS= 'FK5     '         / Coordinate reference frame
LST     =          57000.763 / 15:50:00.763 LST at start (sec)
UTC     =           6018.541 / 01:40:18.541 UTC at start (sec)
......
```

Of all the keywords available in a FITS file the following are the one useful for the reduction of WFI data:

```
OBJECT
EXPTIME
RA
DEC
HIERARCH ESO TEL AIRM START
HIERARCH ESO INS FILT1 NAME
```

where we can see the infamous hierarchical ESO keywords.

Data exploration in the context we are talking about refer to the visualization of the files in the **rawdir** to search for the calibration files and for the science files and to produce ascii lists with selected partitions, e.g. All the deep R band images of the Tempel1 comet. For this we use de Villard's Eclipse command **dfits** and **fitsort**.

```
$dfits /data/reduced/WFI/DeepImpact/2005-07-03/WFI*fits | \
 fitsort OBJECT EXPTIME INS.FILT1.NAME
```

where the backslash character indicate that the command continues in the next line.

An excerpt of the output is displayed below:

```
/data/raw/2005-07-03/WFI.2005-07-03T22:05:25.581.fits  FLAT,SKY   0.8540   MB#445/18_ESO873
/data/raw/2005-07-03/WFI.2005-07-03T22:06:23.775.fits  FLAT,SKY   1.0587   MB#445/18_ESO873
/data/raw/2005-07-03/WFI.2005-07-03T22:07:23.709.fits  FLAT,SKY   1.3219   MB#445/18_ESO873
/data/raw/2005-07-03/WFI.2005-07-03T22:08:24.084.fits  FLAT,SKY   1.6523   MB#445/18_ESO873
/data/raw/2005-07-03/WFI.2005-07-03T22:09:24.058.fits  FLAT,SKY   2.0543   MB#445/18_ESO873
/data/raw/2005-07-03/WFI.2005-07-03T22:12:20.032.fits  OTHER      1.0000   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-03T22:13:36.057.fits  FLAT,SKY   0.7049   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-03T22:14:33.161.fits  FLAT,SKY   0.8666   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-03T22:15:34.236.fits  FLAT,SKY   1.0865   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-03T22:16:31.801.fits  FLAT,SKY   1.3463   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-03T22:17:34.085.fits  FLAT,SKY   1.6953   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-03T22:20:07.346.fits  OTHER      0.9996   BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-03T22:20:57.610.fits  FLAT,SKY   3.5398   BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-03T22:21:59.016.fits  FLAT,SKY   4.4431   BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-03T22:23:04.190.fits  FLAT,SKY   5.6647   BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-03T22:24:08.914.fits  FLAT,SKY   7.2247   BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-03T22:25:14.649.fits  FLAT,SKY   9.2207   BB#Rc/162_ESO844
.........
/data/raw/2005-07-03/WFI.2005-07-04T11:00:29.768.fits  OTHER      1.9919   BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-04T11:04:05.984.fits  OTHER      1.9915   BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-04T11:04:54.708.fits  FLAT,SKY   38.0708  BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-04T11:06:36.506.fits  FLAT,SKY   26.7761  BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-04T11:08:13.443.fits  FLAT,SKY   18.9175  BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-04T11:09:43.019.fits  FLAT,SKY   13.7463  BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-04T11:11:07.746.fits  FLAT,SKY   10.1079  BB#Rc/162_ESO844
/data/raw/2005-07-03/WFI.2005-07-04T11:13:15.695.fits  OTHER      1.9913   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-04T11:14:04.569.fits  FLAT,SKY   3.5191   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-04T11:15:12.904.fits  FLAT,SKY   2.7394   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-04T11:16:22.989.fits  FLAT,SKY   2.1155   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-04T11:17:33.535.fits  FLAT,SKY   1.6355   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-04T11:18:48.560.fits  FLAT,SKY   1.2408   BB#B/123_ESO878
/data/raw/2005-07-03/WFI.2005-07-04T11:21:24.412.fits  OTHER      1.9800   MB#721/25_ESO847
/data/raw/2005-07-03/WFI.2005-07-04T11:22:43.027.fits  FLAT,SKY   2.3621   MB#721/25_ESO847
/data/raw/2005-07-03/WFI.2005-07-04T11:23:46.762.fits  FLAT,SKY   1.8759   MB#721/25_ESO847
/data/raw/2005-07-03/WFI.2005-07-04T11:24:58.457.fits  FLAT,SKY   1.4379   MB#721/25_ESO847
/data/raw/2005-07-03/WFI.2005-07-04T11:26:08.433.fits  FLAT,SKY   1.1108   MB#721/25_ESO847
/data/raw/2005-07-03/WFI.2005-07-04T11:27:23.448.fits  FLAT,SKY   0.8416   MB#721/25_ESO847
/data/raw/2005-07-03/WFI.2005-07-04T11:28:53.815.fits  OTHER      1.9916   MB#445/18_ESO873
/data/raw/2005-07-03/WFI.2005-07-04T11:29:43.239.fits  FLAT,SKY   0.6497   MB#445/18_ESO873
/data/raw/2005-07-03/WFI.2005-07-04T11:30:53.904.fits  FLAT,SKY   0.5014   MB#445/18_ESO873
```

One can then use the above command together with the unix *grep* command to select appropriate files, e.g.

```
$dfits /data/reduced/WFI/DeepImpact/2005-07-03/WFI*fits | fitsort OBJECT EXPTIME \
 INS.FILT1.NAME | grep FLAT,SKY | grep Rc | awk '{printf "%s\n", $1}' >ffsky_Rc.list
```

```
$cat ffsky_Rc.list
/data/raw/2005-07-03/WFI.2005-07-03T22:20:57.610.fits
/data/raw/2005-07-03/WFI.2005-07-03T22:21:59.016.fits
/data/raw/2005-07-03/WFI.2005-07-03T22:23:04.190.fits
/data/raw/2005-07-03/WFI.2005-07-03T22:24:08.914.fits
/data/raw/2005-07-03/WFI.2005-07-03T22:25:14.649.fits
/data/raw/2005-07-03/WFI.2005-07-04T11:04:54.708.fits
/data/raw/2005-07-03/WFI.2005-07-04T11:06:36.506.fits
/data/raw/2005-07-03/WFI.2005-07-04T11:08:13.443.fits
/data/raw/2005-07-03/WFI.2005-07-04T11:09:43.019.fits
/data/raw/2005-07-03/WFI.2005-07-04T11:11:07.746.fits
```

Step 1: Create ascii files with the bias frames, bad pixel map frames, dome flat frames, and sky flat frames:

$cat bias.list
/data/raw/2005-07-03/WFI_Dark.1.fits
/data/raw/2005-07-03/WFI_Dark.2.fits
/data/raw/2005-07-03/WFI_Dark.3.fits
/data/raw/2005-07-03/WFI_Dark.4.fits
/data/raw/2005-07-03/WFI_Dark.5.fits
/data/raw/2005-07-03/WFI_Dark.6.fits
/data/raw/2005-07-03/WFI_Dark.7.fits
/data/raw/2005-07-03/WFI_Dark.8.fits
/data/raw/2005-07-03/WFI_Dark.9.fits

$cat bpm.list
/data/raw/2005-07-03/WFI_FlatLampIma.fits
/data/raw/2005-07-03/WFI_FlatLampIma.1.fits
/data/raw/2005-07-03/WFI_FlatLampIma.2.fits
/data/raw/2005-07-03/WFI_FlatLampIma.3.fits
/data/raw/2005-07-03/WFI_FlatLampIma.4.fits
/data/raw/2005-07-03/WFI_FlatLampIma.5.fits
/data/raw/2005-07-03/WFI_FlatLampIma.6.fits
/data/raw/2005-07-03/WFI_FlatLampIma.7.fits

$**cat domeff_Rc.list**
/data/raw/2005-07-03/WFI_FlatDomeIma.fits
/data/raw/2005-07-03/WFI_FlatDomeIma.1.fits
/data/raw/2005-07-03/WFI_FlatDomeIma.2.fits
/data/raw/2005-07-05/WFI_FlatDomeIma.6.fits
/data/raw/2005-07-05/WFI_FlatDomeIma.7.fits
/data/raw/2005-07-05/WFI_FlatDomeIma.8.fits

$**cat skyff_Rc.list**
/data/raw/2005-07-03/WFI_FlatSkyIma.fits
/data/raw/2005-07-03/WFI_FlatSkyIma.1.fits
/data/raw/2005-07-03/WFI_FlatSkyIma.2.fits
/data/raw/2005-07-03/WFI_FlatSkyIma.3.fits
/data/raw/2005-07-03/WFI_FlatSkyIma.4.fits

In the above example we have provided a list with raw frames for all types of calibrations. In practice the user will probably skip the bad pixel masks which can be obtained from the WFI web pages. They come in single extension FITS files:

WFI_BPM_01.fits
WFI_BPM_02.fits
WFI_BPM_03.fits
WFI_BPM_04.fits
WFI_BPM_05.fits
WFI_BPM_06.fits
WFI_BPM_07.fits
WFI_BPM_08.fits

They have the value 1 if the pixel is OK and 0 if it is bad. ***If the user prefer not to create the BPM files him/herself then the above files should be present in the directory where the data is being reduced. A set is distributed in the ins/mef/static_calib directory of this software distribution.***

Step 2: Produce the calibration data the user should execute the command:

```
$mefpp --bias bias.list --biassave --dome domeff_Rc.list --domesave \
    --skyff skyff_Rc.list --skysave
```

At the end of this command, which can take a long time to execute, the program will have written to disk the files:

WFI_MB_1x1.fits
WFI_DOMEFF_BB_Rc_162_ESO844_1x1.fits
WFI_SKYFF_BB_Rc_162_ESO844_1x1.fits
WFI_MFF_BB_Rc_162_ESO844_1x1.fits

These are MEF files in floating point with a size of 268MBytes each.

Step 3: Reduce the science frames by writing their names in a file, e.g. tempel1_Rc.list

```
$dfits  /data/raw/2005-07-03/WFI*fits | fitsort OBJECT EXPTIME INS.FILT1.NAME |  \
 grep Tempel1 | grep Rc | grep 299 | awk '{printf "%s\n",$1}' >Tempel1_Rc.list
$cat tempel1_Rc.list
/data/raw/2005-07-03/WFI.2005-07-03T23:07:50.699.fits
/data/raw/2005-07-03/WFI.2005-07-03T23:16:54.380.fits
/data/raw/2005-07-03/WFI.2005-07-03T23:23:58.061.fits
/data/raw/2005-07-03/WFI.2005-07-03T23:38:16.955.fits
/data/raw/2005-07-03/WFI.2005-07-03T23:45:17.787.fits
/data/raw/2005-07-03/WFI.2005-07-03T23:52:22.338.fits
/data/raw/2005-07-03/WFI.2005-07-04T00:01:44.400.fits
/data/raw/2005-07-03/WFI.2005-07-04T00:09:40.275.fits
/data/raw/2005-07-03/WFI.2005-07-04T00:17:19.940.fits
/data/raw/2005-07-03/WFI.2005-07-04T00:24:33.131.fits
/data/raw/2005-07-03/WFI.2005-07-04T00:38:49.766.fits
/data/raw/2005-07-03/WFI.2005-07-04T00:47:10.873.fits
/data/raw/2005-07-03/WFI.2005-07-04T00:54:27.265.fits
/data/raw/2005-07-03/WFI.2005-07-04T01:12:37.306.fits
/data/raw/2005-07-03/WFI.2005-07-04T01:19:45.878.fits
/data/raw/2005-07-03/WFI.2005-07-04T01:29:00.109.fits
/data/raw/2005-07-03/WFI.2005-07-04T01:36:38.283.fits
/data/raw/2005-07-03/WFI.2005-07-04T01:44:04.817.fits
/data/raw/2005-07-03/WFI.2005-07-04T03:08:30.113.fits
/data/raw/2005-07-03/WFI.2005-07-04T03:15:39.525.fits
/data/raw/2005-07-03/WFI.2005-07-04T03:23:11.508.fits
/data/raw/2005-07-03/WFI.2005-07-04T03:31:07.894.fits
/data/raw/2005-07-03/WFI.2005-07-04T04:15:16.351.fits
/data/raw/2005-07-03/WFI.2005-07-04T04:22:11.592.fits
```

Then execute the command

```
$mefpp --bias WFI_MB_1x1.fits --mff WFI_SKYFF_BB_Rc_162_ESO844_1x1.fits \
        --science tempel1_Rc.list
```

After some time the program will stop having writen to disk, in **workdir**, the files
WFI.2005-07-03T23:07:50.699_pre.fits
WFI.2005-07-03T23:16:54.380_pre.fits
WFI.2005-07-03T23:23:58.061_pre.fits
WFI.2005-07-03T23:38:16.955_pre.fits

WFI.2005-07-03T23:45:17.787_pre.fits
WFI.2005-07-03T23:52:22.338_pre.fits
WFI.2005-07-04T00:01:44.400_pre.fits
WFI.2005-07-04T00:09:40.275_pre.fits
WFI.2005-07-04T00:17:19.940_pre.fits
WFI.2005-07-04T00:24:33.131_pre.fits
WFI.2005-07-04T00:38:49.766_pre.fits
WFI.2005-07-04T00:47:10.873_pre.fits
WFI.2005-07-04T00:54:27.265_pre.fits
WFI.2005-07-04T01:12:37.306_pre.fits
WFI.2005-07-04T01:19:45.878_pre.fits
WFI.2005-07-04T01:29:00.109_pre.fits
WFI.2005-07-04T01:36:38.283_pre.fits
WFI.2005-07-04T01:44:04.817_pre.fits
WFI.2005-07-04T03:08:30.113_pre.fits
WFI.2005-07-04T03:15:39.525_pre.fits
WFI.2005-07-04T03:23:11.508_pre.fits
WFI.2005-07-04T03:31:07.894_pre.fits
WFI.2005-07-04T04:15:16.351_pre.fits
WFI.2005-07-04T04:22:11.592_pre.fits

These are MEF files 268Mbytes each. They have their CRVALi and CRPIXi keywords corrected, overscan subtracted, bias subtracted, and flat fielded. The overcan is subtracted line by line. We have chosen above to use the sky flat field instead of the master flat field because dome flats for WFI are not very good.

README

Wide Field Imager Reduction recipes

(F. Selman, La Silla 17 February 2005)

The recipes found in this package use the originals written by N. Devillard as starting point.

They have been **extensively** modified to work with WFI MEF files directly, both as input and output. I decided to keep the images always within a MEF file which is their logical place, at least in the context of a primary processing. For example, the flat field routines now keep track of the median in each of the extension and save the value in the respective BSCALE keywords. This save us from the need to do a gain harmonization of the chips later on, as the flat field give us that directly if we take the precaution of normalizing the flat field frames with a single number for all the extensions. Thus, the user should work in ADUs until after the frames have been flat fielded.

The basic modus operandi is to go extension by extension applying the already proven eclipse routines.

Version 1.5 of these recipes should deal correctly with the following issues:

- Input and output to MEF files
- Proper treatment of overscan correction, subtracting a line rather than a single number, as the overscan varies with y.
- Proper treatment of the BSCALE and BZERO keywords.
- BPM creation, and inclusion in the processing
- Proper treatment of the CRVALi, and CRPIXi keywords, specially after trimming.
- Determination of WCS and inclusion in headers

These recipes are intended as efficient and simple to use, voiding the complexities of a pipeline.
PATCHES:

> The current distribution includes a patch to the eclipse routine src/unix/comm.c.

This routine has calls to vafprintf which
core dumps when run with gcc in ppc. This
patch is not needed in x86 Linux or in
Mac OS X. It only bombs in Linux PPC.

F. Selman

REAME FROM ORIGINAL VERSION:

The recipes found in this directory have been written in the context of the BeoWFI project, aimed at providing a data reduction facility for the WFI instrument based on a Beowulf cluster.

The recipe functionalities are drawn from the eclipse library. The recipe code has been written by N. Devillard, based on inputs from Aniello Grado, Andreas Wicenec, Stefan Bogun, Miguel Albrecht, Peter Quinn. The Beowulf demonstration cluster at ESO is maintained by Dieter Suchar and Andreas Wicenec.

N. Devillard
Wed Oct 25 13:14:02 CEST 2000

TODO (and never done...)
(written 17 February 2005)

-Reading from a PAF file recipe parameters,
and instrument parameters.

-Proper identification of recipes:

    REC MEF BIAS
    REC MEF BPM
    REC MEF OVERSCAN
    REC MEF DOMEFF
    REC MEF SKYFF
    REC MEF MASTERFF

-Clean up: using qfits as far as possible

-Proper handling of writing to files:
see the pixdump example and modify/integrate
into a routine to dump headers and pixels
to MEF files.

-Further separation into header creation and
file writing

-Find proper way to deal with command line options,
and decide on weather we will have a single large
pipe-like processing, or many atomic units.

-Add option to use a "normalized" flat, that is,
a flat that only takes out the pixel-to-pixel
variations.

-Write man page

-Test parameterers for BPM creation

-Test Flat field creation and write script
to monitor stability of flatfield for
keyband pixels:

    Dome/Dome
    Sky/Sky
    Dome/Sky

- Add a linearity correction section

- Add a shutter error correction section

- Add proper WCS keyword handling

- Add registering/warping routine

- Create a few coulour pictures

- Test with OmegaCAM data.